Graphing Rational Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Reading graphs**

The graph of $f(x) = \frac{x+2}{x-3}$ is shown. From the graph and the formula, state the asymptotes and intercepts.

Vertical asymptote $x = 3$; horizontal asymptote $y = 1$; intercepts $(-2, 0)$ and $\left(0, -\frac{2}{3}\right)$.

2 **[Foundational]** • **Target: Reading graphs**

The function $g(x) = \frac{1}{x-2} + 3$ is a transformation of $y = \frac{1}{x}$.

a Describe the transformation. Shift right 2 units and up 3 units.

b State the asymptotes of g. Vertical $x = 2$; horizontal $y = 3$.

c State the domain and range. Domain $x \neq 2$; range $y \neq 3$.

(a) Shift right 2 units and up 3 units.; (b) Vertical $x = 2$; horizontal $y = 3$.; (c) Domain $x \neq 2$; range $y \neq 3$.

3 **[Foundational]** • **Target: Sign analysis**

Let $f(x) = \frac{x-1}{(x+2)(x-4)}$. Using a sign chart with critical values $x = -2, 1, 4$, determine the intervals on which $f(x) > 0$.

Testing each interval: $(-\infty, -2)$ negative; $(-2, 1)$ positive; $(1, 4)$ negative; $(4, \infty)$ positive. So $f(x) > 0$ on $(-2, 1) \cup (4, \infty)$.

4 **[Foundational]** • **Target: Sign analysis**

Solve the rational inequality $\frac{x+3}{x-2} \leq 0$.

Critical values $x = -3$ (zero) and $x = 2$ (undefined). The quotient is negative between them: $[-3, 2)$ — including -3, excluding 2.

5 **[Foundational]** • **Target: Sign analysis**

Sketch-planning: for $h(x) = \frac{2x}{x^2 - 1}$, state the asymptotes and the symmetry of the graph.

Vertical asymptotes $x = 1$ and $x = -1$; horizontal asymptote $y = 0$. Since $h(-x) = -h(x)$, the function is odd — symmetric about the origin.

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6 [Foundational] • Target: Sketching a complete graph

For $f(x) = \frac{x^2-1}{x^2-4}$, find all intercepts and asymptotes, then describe the graph's behaviour near each vertical asymptote.

x-intercepts: $x = \pm 1$. y-intercept: $f(0) = \frac{-1}{-4} = \frac{1}{4}$. Vertical asymptotes $x = \pm 2$; horizontal asymptote $y = 1$ (equal degrees). Near $x = 2$: $f \to +\infty$ as $x \to 2^+$ and $f \to -\infty$ as $x \to 2^-$. Near $x = -2$: by symmetry (the function is even), $f \to -\infty$ as $x \to -2^+$ and $f \to +\infty$ as $x \to -2^-$.

7 [Foundational] • Target: Sketching a complete graph

Sketch-planning: the function $g(x) = \frac{3}{x-1} - 2$ has a vertical asymptote at $x = 1$ and horizontal asymptote $y = -2$. Find its x- and y-intercepts.

y-intercept: $g(0) = \frac{3}{-1} - 2 = -5$. x-intercept: $0 = \frac{3}{x-1} - 2$ gives $\frac{3}{x-1} = 2$, so $x - 1 = \frac{3}{2}$, $x = \frac{5}{2}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: More rational inequalities

Solve $\frac{(x-2)(x+1)}{x-4} \geq 0$.

Critical values $x = -1, 2, 4$. Sign chart: negative on $(-\infty, -1)$, positive on $(-1, 2)$, negative on $(2, 4)$, positive on $(4, \infty)$. Solution (including zeros, excluding the undefined point): $[-1, 2] \cup (4, \infty)$.

9 [Intermediate] • Target: More rational inequalities

Solve $\frac{2}{x+3} > 1$.

Rewrite: $\frac{2}{x+3} - 1 > 0 \Rightarrow \frac{2-(x+3)}{x+3} > 0 \Rightarrow \frac{-x-1}{x+3} > 0 \Rightarrow \frac{x+1}{x+3} < 0$. Critical values $-3, -1$: solution $(-3, -1)$.

10 [Intermediate] • Target: Building a rational function from a description

Construct a rational function with vertical asymptote $x = 2$, horizontal asymptote $y = 1$, and a hole at $x = -3$.

Start with a base of $\frac{x}{x-2}$ (horizontal asymptote $y = 1$, vertical asymptote $x = 2$), then introduce a common factor $(x + 3)$ in both numerator and denominator: $f(x) = \frac{x(x+3)}{(x-2)(x+3)}$.

11 [Intermediate] • Target: Graphing with a slant asymptote

For $f(x) = \frac{x^2-4}{x-1}$, find the slant asymptote and the vertical asymptote, and use them together with intercepts to describe the overall shape of the graph.

Dividing: $x^2 - 4 = (x-1)(x+1) - 3$, so $f(x) = x + 1 - \frac{3}{x-1}$. Slant asymptote: $y = x + 1$. Vertical asymptote: $x = 1$. x-intercepts: $x = \pm 2$. As $x \to \pm\infty$, the graph hugs the line $y = x + 1$, while near $x = 1$ it shoots off to $\pm\infty$.

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12 **[Intermediate]** • **Target: Full graph analysis: putting it all together**

For $f(x) = \frac{2x^2 - 2}{x^2 - 9}$, find all intercepts, all asymptotes, and describe the symmetry of the graph (even, odd, or neither).

x-intercepts: $2x^2 - 2 = 0 \Rightarrow x = \pm 1$. y-intercept: $f(0) = \frac{-2}{-9} = \frac{2}{9}$. Vertical asymptotes: $x = \pm 3$. Horizontal asymptote (equal degrees): $y = 2$. Symmetry: $f(-x) = \frac{2x^2 - 2}{x^2 - 9} = f(x)$, so the function is EVEN, symmetric about the y-axis.

13 **[Intermediate]** • **Target: A compound rational inequality**

Solve $\frac{x^2 - 1}{x - 2} \leq 0$.

Factor numerator: $\frac{(x-1)(x+1)}{x-2} \leq 0$. Critical values $-1, 1, 2$. Sign chart: negative on $(-\infty, -1)$, positive on $(-1, 1)$, negative on $(1, 2)$, positive on $(2, \infty)$. Solution (including zeros, excluding undefined point): $(-\infty, -1] \cup [1, 2)$.

14 **[Intermediate]** • **Target: Graphing near a hole**

For $f(x) = \frac{x^2 - 9}{x - 3}$, describe how the graph looks immediately around $x = 3$, and find the coordinates of the hole.

Simplify: $f(x) = x + 3$ for $x \neq 3$. The graph is just the line $y = x + 3$ with a single point removed at $x = 3$: the hole is at $(3, 6)$ — an open circle on an otherwise straight line.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Graphing near a hole** • **Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations**

Solve $\frac{3}{x+1} \geq \frac{2}{x-1}$.

Move all terms to one side: $\frac{3}{x+1} - \frac{2}{x-1} \geq 0 \Rightarrow \frac{3(x-1) - 2(x+1)}{(x+1)(x-1)} \geq 0 \Rightarrow \frac{x-5}{(x+1)(x-1)} \geq 0$. Critical values $-1, 1, 5$. Sign chart: negative on $(-\infty, -1)$, positive on $(-1, 1)$, negative on $(1, 5)$, positive on $(5, \infty)$. Solution: $(-1, 1) \cup [5, \infty)$.

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16 [Challenging] • Target: Graphing near a hole • Cross-Lesson: 4.01 Circles and Parabolas

A rectangular garden's cost per square meter to fence is modeled by $C(x) = \frac{200}{x} + 8x$, where x is the garden's width in meters. Find the vertical asymptote and horizontal/slant asymptote of this function, and interpret what the vertical asymptote means physically (why $x = 0$ isn't a realistic input).

Vertical asymptote: $x = 0$. As $x \to 0^+$, $C(x) \to \infty$ — a garden of near-zero width would require infinite fencing cost per unit length, which isn't physically meaningful; this confirms $x = 0$ must be excluded from the domain. There is no horizontal asymptote (the $8x$ term grows without bound), and the function behaves like $y = 8x$ for large x (a slant-like trend, though not a strict polynomial slant asymptote since $\frac{200}{x} \to 0$).

17 [Challenging] • Target: Graphing near a hole • Cross-Lesson: 4.02 Ellipses

State the domain of $f(x) = \frac{\sqrt{x}}{x-4}$.

Need $x \geq 0$ (for the square root) and $x \neq 4$ (denominator). Domain: $[0, 4) \cup (4, \infty)$.

18 [Challenging] • Target: Building a rational function from a description • Cross-Lesson: 4.03 Hyperbolas

For $f(x) = \frac{(x+2)(x+1)}{(x+1)(x+3)}$, identify the hole and vertical asymptote.

Final Answer: Hole at $x = -1$, vertical asymptote at $x = -3$. The factor $(x + 1)$ cancels, creating a hole at $x = -1$; the remaining zero of the denominator, $x = -3$, gives the vertical asymptote.

19 [Challenging] • Target: Graphing with a slant asymptote • Cross-Lesson: 5.01 Vectors: Components and Operations

For $f(x) = \frac{(x+1)(x+2)}{(x+2)(x+5)}$, identify the hole and vertical asymptote.

Final Answer: Hole at $x = -2$, vertical asymptote at $x = -5$. The factor $(x + 2)$ cancels, creating a hole at $x = -2$; the remaining zero of the denominator, $x = -5$, gives the vertical asymptote.

20 [Challenging] • Target: Full graph analysis: putting it all together • Cross-Lesson: 5.02 The Dot Product

For $f(x) = \frac{(x-0)(x+1)}{(x+1)(x+3)}$, identify the hole and vertical asymptote.

Final Answer: Hole at $x = -1$, vertical asymptote at $x = -3$. The factor $(x + 1)$ cancels, creating a hole at $x = -1$; the remaining zero of the denominator, $x = -3$, gives the vertical asymptote.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiplicity of Zeros and Sign Analysis

If a factor $(x - c)$ in the denominator of a rational function has odd multiplicity (e.g. $(x - c)^1$), how does the graph behave near the asymptote $x = c$?

a. The graph goes in opposite directions ($+\infty$ on one side and $-\infty$ on the other) ✓

b. The graph goes in the same direction ($+\infty$ on both sides)

c. The graph crosses the axis

d. A hole is formed

Odd multiplicity produces a sign change across the asymptote, sending one side toward $+\infty$ and the other toward $-\infty$.

22 [Foundational] • Target: Multiplicity of Zeros and Sign Analysis

If the factor in the denominator has even multiplicity $(x - c)^2$, how does the graph behave near $x = c$?

a. The graph goes in the same direction on both sides (both $+\infty$ or both $-\infty$) ✓

b. The graph goes in opposite directions

c. The graph crosses the asymptote

d. A corner is formed

Even multiplicity maintains the same sign on both sides of c, sending both branches toward the same infinity.

23 [Foundational] • Target: Multiplicity of Zeros and Sign Analysis

What is the sign behavior of $f(x) = \frac{1}{(x-2)^2}$ on both sides of $x = 2$?

a. Positive ($+\infty$) on both sides ✓

b. Negative on left, positive on right

c. Negative on both sides

d. Undefined

Since $(x - 2)^2 > 0$ for all $x \neq 2$, the output is strictly positive, approaching $+\infty$ from both sides.

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24 [Foundational] • Target: Constructing Sign Charts

Which values partition the number line into test intervals when making a sign chart for a rational function?

a. All real zeros of the numerator and all real zeros of the denominator ✓

b. Only the zeros of the denominator

c. Only the zeros of the numerator

d. Only the *y*-intercept

A rational function can change signs only at the zeros of its numerator (*x*-intercepts) and the zeros of its denominator (vertical asymptotes).

25 [Foundational] • Target: Constructing Sign Charts

Find the solution set of $\frac{x-3}{x+1} > 0$.

a. $(-\infty, -1) \cup (3, \infty)$ ✓

b. $(-1, 3)$

c. $[3, \infty)$

d. $(-\infty, 3)$

Critical points at $x = -1, 3$. For $x < -1$: $(-)/(-) = +$. For $-1 < x < 3$: $(-)/(+) = -$. For $x > 3$: $(+)/(+) = +$. Thus $(-\infty, -1) \cup (3, \infty)$.

26 [Foundational] • Target: Constructing Sign Charts

Solve the inequality $\frac{x^2-4}{x} \leq 0$.

a. $(-\infty, -2] \cup (0, 2]$ ✓

b. $[-2, 0) \cup [2, \infty)$

c. $[-2, 2]$

d. $(-\infty, 0)$

Key values: $-2, 0, 2$. Test: $x < -2 \Rightarrow -$; $-2 \leq x < 0 \Rightarrow +$; $0 < x \leq 2 \Rightarrow -$; $x > 2 \Rightarrow +$. Since $x = 0$ is in denominator, interval is $(-\infty, -2] \cup (0, 2]$.

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27 [Foundational] • Target: Sketching Complete Curves

Which of the following is the complete set of features needed to accurately sketch a rational function?

a. Domain, holes, vertical/horizontal asymptotes, x- and y-intercepts, and sign behavior ✓

b. Only the degree and the leading coefficient

c. Only the y-intercept and one point

d. Only the vertical asymptotes

Accurate sketching requires determining all discontinuities, asymptotes, intercepts, and branch behaviors between asymptotes.

28 [Foundational] • Target: Sketching Complete Curves

Sketch $f(x) = \frac{2x}{x-1}$. What is the value of $f(2)$ and $f(-1)$?

a. $f(2) = 4$ and $f(-1) = 1$ ✓

b. $f(2) = 2$ and $f(-1) = -1$

c. $f(2) = 4$ and $f(-1) = -1$

d. $f(2) = 1$ and $f(-1) = 2$

$f(2) = \frac{4}{1} = 4$. $f(-1) = \frac{-2}{-2} = 1$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Sketching Complete Curves

In $f(x) = \frac{2x}{x-1}$, as $x \to 1^+$, what is the limit of $f(x)$?

a. $+\infty$ ✓

b. $-\infty$

c. 2

d. 0

As $x \to 1^+$, numerator $\to 2 > 0$ and denominator $\to 0^+$, so $f(x) \to +\infty$.

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30 [Intermediate] • Target: Writing Rational Functions from Features

Write a rational function with vertical asymptotes at $x = -2$ and $x = 3$, horizontal asymptote $y = 0$, and x-intercept $(1, 0)$.

a. $f(x) = \frac{x-1}{(x+2)(x-3)}$ ✓

b. $f(x) = \frac{(x+2)(x-3)}{x-1}$

c. $f(x) = \frac{x-1}{x^2+1}$

d. $f(x) = \frac{2x-1}{(x+2)(x-3)}$

Numerator has factor $(x-1)$ to give x-intercept 1. Denominator has factors $(x+2)(x-3)$. Since $\deg(\text{Num}) = 1 < \deg(\text{Den}) = 2$, horizontal asymptote is $y = 0$.

31 [Intermediate] • Target: Writing Rational Functions from Features

Write a rational function with vertical asymptote $x = 4$, hole at $x = -1$, horizontal asymptote $y = 2$, and y-intercept $(0, -1/2)$.

a. $f(x) = \frac{2(x+1)(x-1)}{(x+1)(x-4)}$ ✓

b. $f(x) = \frac{2x+1}{x-4}$

c. $f(x) = \frac{(x+1)}{(x+1)(x-4)}$

d. $f(x) = \frac{2(x-4)}{x+1}$

Hole at $-1 \Rightarrow (x+1)$ in both. Vertical asymptote at $4 \Rightarrow (x-4)$ in denominator. Ratio of leading coefficients $= 2$. $f(0) = \frac{2(1)(-1)}{(1)(-4)} = \frac{-2}{-4} = \frac{1}{2}$; with $(x-1)$ in numerator gives $f(0) = 1/2$ or $-1/2$.

32 [Intermediate] • Target: Writing Rational Functions from Features

What is the minimum degree of the denominator of a rational function with two distinct vertical asymptotes?

a. 2 ✓

b. 1

c. 3

d. 4

Each distinct vertical asymptote requires at least one distinct linear factor in the denominator, so degree ≥ 2.

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33 [Intermediate] • Target: Writing Rational Functions from Features

For $f(x) = \frac{(x-4)(x+6)}{(x+6)(x+3)}$, identify the hole and vertical asymptote.

a. Hole at $x = -3$, vertical asymptote at $x = -6$

b. Vertical asymptotes at $x = -6$ and $x = -3$, no hole

c. Hole at $x = 4$, vertical asymptote at $x = -3$

d. **Hole at $x = -6$, vertical asymptote at $x = -3$** ✓

The factor $(x + 6)$ cancels, creating a hole at $x = -6$; the remaining zero of the denominator, $x = -3$, gives the vertical asymptote.

34 [Intermediate] • Target: Reading graphs

For $f(x) = \frac{(x-5)(x+5)}{(x+5)(x+3)}$, identify the hole and vertical asymptote.

a. Hole at $x = 5$, vertical asymptote at $x = -3$

b. **Hole at $x = -5$, vertical asymptote at $x = -3$** ✓

c. Hole at $x = -3$, vertical asymptote at $x = -5$

d. Vertical asymptotes at $x = -5$ and $x = -3$, no hole

The factor $(x + 5)$ cancels, creating a hole at $x = -5$; the remaining zero of the denominator, $x = -3$, gives the vertical asymptote.

35 [Intermediate] • Target: Sign analysis

For $f(x) = \frac{(x-3)(x+2)}{(x+2)(x+4)}$, identify the hole and vertical asymptote.

a. Hole at $x = 3$, vertical asymptote at $x = -4$

b. Hole at $x = -4$, vertical asymptote at $x = -2$

c. **Hole at $x = -2$, vertical asymptote at $x = -4$** ✓

d. Vertical asymptotes at $x = -2$ and $x = -4$, no hole

The factor $(x + 2)$ cancels, creating a hole at $x = -2$; the remaining zero of the denominator, $x = -4$, gives the vertical asymptote.

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36 [Intermediate] • Target: Sketching a complete graph

For $f(x) = \frac{(x-5)(x+6)}{(x+6)(x+4)}$, identify the hole and vertical asymptote.

a. Hole at $x = 5$, vertical asymptote at $x = -4$

b. Hole at $x = -4$, vertical asymptote at $x = -6$

c. **Hole at $x = -6$, vertical asymptote at $x = -4$** ✓

d. Vertical asymptotes at $x = -6$ and $x = -4$, no hole

The factor $(x + 6)$ cancels, creating a hole at $x = -6$; the remaining zero of the denominator, $x = -4$, gives the vertical asymptote.

37 [Intermediate] • Target: More rational inequalities

For $f(x) = \frac{(x-1)(x+6)}{(x+6)(x+4)}$, identify the hole and vertical asymptote.

a. Hole at $x = 1$, vertical asymptote at $x = -4$

b. **Hole at $x = -6$, vertical asymptote at $x = -4$** ✓

c. Vertical asymptotes at $x = -6$ and $x = -4$, no hole

d. Hole at $x = -4$, vertical asymptote at $x = -6$

The factor $(x + 6)$ cancels, creating a hole at $x = -6$; the remaining zero of the denominator, $x = -4$, gives the vertical asymptote.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Building a rational function from a description • Cross-Lesson: 4.03 Hyperbolas

For $f(x) = \frac{(x+5)(x+3)}{(x+3)(x-0)}$, identify the hole and vertical asymptote.

a. Hole at $x = 0$, vertical asymptote at $x = -3$

b. Vertical asymptotes at $x = -3$ and $x = 0$, no hole

c. Hole at $x = -5$, vertical asymptote at $x = 0$

d. **Hole at $x = -3$, vertical asymptote at $x = 0$** ✓

The factor $(x + 3)$ cancels, creating a hole at $x = -3$; the remaining zero of the denominator, $x = 0$, gives the vertical asymptote.

39 [Challenging] • Target: Graphing with a slant asymptote • Cross-Lesson: 5.01 Vectors: Components and Operations

For $f(x) = \frac{(x-5)(x+1)}{(x+1)(x-2)}$, identify the hole and vertical asymptote.

a. Vertical asymptotes at $x = -1$ and $x = 2$, no hole

b. Hole at $x = 5$, vertical asymptote at $x = 2$

c. **Hole at $x = -1$, vertical asymptote at $x = 2$** ✓

d. Hole at $x = 2$, vertical asymptote at $x = -1$

The factor $(x + 1)$ cancels, creating a hole at $x = -1$; the remaining zero of the denominator, $x = 2$, gives the vertical asymptote.

40 [Challenging] • Target: Full graph analysis: putting it all together • Cross-Lesson: 5.02 The Dot Product

For $f(x) = \frac{(x+6)(x+2)}{(x+2)(x-0)}$, identify the hole and vertical asymptote.

a. Vertical asymptotes at $x = -2$ and $x = 0$, no hole

b. Hole at $x = -6$, vertical asymptote at $x = 0$

c. **Hole at $x = -2$, vertical asymptote at $x = 0$** ✓

d. Hole at $x = 0$, vertical asymptote at $x = -2$

The factor $(x + 2)$ cancels, creating a hole at $x = -2$; the remaining zero of the denominator, $x = 0$, gives the vertical asymptote.

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41 [Challenging] • Target: A compound rational inequality • Cross-Lesson: 5.03 Parametric Equations

For $f(x) = \frac{(x-3)(x-0)}{(x-0)(x+3)}$, identify the hole and vertical asymptote.

a. **Hole at $x = 0$, vertical asymptote at $x = -3$** ✓

b. Vertical asymptotes at $x = 0$ and $x = -3$, no hole

c. Hole at $x = 3$, vertical asymptote at $x = -3$

d. Hole at $x = -3$, vertical asymptote at $x = 0$

The factor $(x - 0)$ cancels, creating a hole at $x = 0$; the remaining zero of the denominator, $x = -3$, gives the vertical asymptote.

42 [Challenging] • Target: Graphing near a hole • Cross-Lesson: 6.01 Polar Coordinates and Conversions

For $f(x) = \frac{(x+6)(x-1)}{(x-1)(x-4)}$, identify the hole and vertical asymptote.

a. **Hole at $x = 1$, vertical asymptote at $x = 4$** ✓

b. Hole at $x = -6$, vertical asymptote at $x = 4$

c. Vertical asymptotes at $x = 1$ and $x = 4$, no hole

d. Hole at $x = 4$, vertical asymptote at $x = 1$

The factor $(x - 1)$ cancels, creating a hole at $x = 1$; the remaining zero of the denominator, $x = 4$, gives the vertical asymptote.

43 [Challenging] • Target: Multiplicity of Zeros and Sign Analysis • Cross-Lesson: 6.02 Polar Equations and Graphs

For $f(x) = \frac{(x-1)(x-3)}{(x-3)(x-0)}$, identify the hole and vertical asymptote.

a. Hole at $x = 0$, vertical asymptote at $x = 3$

b. Hole at $x = 1$, vertical asymptote at $x = 0$

c. Vertical asymptotes at $x = 3$ and $x = 0$, no hole

d. **Hole at $x = 3$, vertical asymptote at $x = 0$** ✓

The factor $(x - 3)$ cancels, creating a hole at $x = 3$; the remaining zero of the denominator, $x = 0$, gives the vertical asymptote.

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44 [Challenging] • Target: Constructing Sign Charts • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

For $f(x) = \frac{(x-0)(x-3)}{(x-3)(x-5)}$, identify the hole and vertical asymptote.

a. Hole at $x = 5$, vertical asymptote at $x = 3$

b. Hole at $x = 0$, vertical asymptote at $x = 5$

c. Vertical asymptotes at $x = 3$ and $x = 5$, no hole

d. **Hole at $x = 3$, vertical asymptote at $x = 5$** ✓

The factor $(x - 3)$ cancels, creating a hole at $x = 3$; the remaining zero of the denominator, $x = 5$, gives the vertical asymptote.

45 [Challenging] • Target: Sketching Complete Curves • Cross-Lesson: 7.02 Logarithms and Their Properties

For $f(x) = \frac{(x+4)(x-4)}{(x-4)(x-1)}$, identify the hole and vertical asymptote.

a. **Hole at $x = 4$, vertical asymptote at $x = 1$** ✓

b. Vertical asymptotes at $x = 4$ and $x = 1$, no hole

c. Hole at $x = 1$, vertical asymptote at $x = 4$

d. Hole at $x = -4$, vertical asymptote at $x = 1$

The factor $(x - 4)$ cancels, creating a hole at $x = 4$; the remaining zero of the denominator, $x = 1$, gives the vertical asymptote.

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